

# MPC-PiNNs technique applied to Quadrotors

Maiorana, Flavio. Bugli, Eugenio. Di Marzo, Gabriele

February 11, 2025

## Contents

|          |                                        |           |
|----------|----------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                    | <b>3</b>  |
| <b>2</b> | <b>Models</b>                          | <b>3</b>  |
| 2.1      | Unicycle kinematic model . . . . .     | 3         |
| 2.2      | Quadrotor dynamic model . . . . .      | 4         |
| <b>3</b> | <b>Model Predictive Control</b>        | <b>4</b>  |
| 3.1      | Cost function . . . . .                | 5         |
| 3.2      | Constraints . . . . .                  | 6         |
| 3.3      | Unicycle . . . . .                     | 6         |
| 3.4      | Quadrotor . . . . .                    | 7         |
| <b>4</b> | <b>Dataset</b>                         | <b>8</b>  |
| <b>5</b> | <b>Neural models</b>                   | <b>10</b> |
| 5.0.1    | Architecture . . . . .                 | 10        |
| 5.1      | Standard neural network . . . . .      | 10        |
| 5.1.1    | Training scheme . . . . .              | 10        |
| 5.1.2    | Consideration of the Dataset . . . . . | 11        |
| 5.2      | Unicycle Neural model . . . . .        | 11        |
| 5.2.1    | Architecture . . . . .                 | 11        |
| 5.2.2    | Dataset . . . . .                      | 11        |
| 5.2.3    | Training . . . . .                     | 12        |
| 5.3      | Quadrotor Neural model . . . . .       | 12        |
| 5.3.1    | Architecture . . . . .                 | 12        |
| 5.3.2    | Dataset . . . . .                      | 13        |
| 5.3.3    | Training . . . . .                     | 13        |

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| <b>6</b> | <b>Physical Informed Models</b>                            | <b>13</b> |
| 6.1      | Physics-Informed Neural Network: Introduction . . . . .    | 13        |
| 6.2      | Physics-Informed Neural Network with Control . . . . .     | 14        |
| 6.2.1    | Dataset requirements, generation and constraints . . . . . | 15        |
| 6.2.2    | Loss definition and training scheme . . . . .              | 16        |
| <b>7</b> | <b>Results</b>                                             | <b>17</b> |
| 7.1      | Derivative Learning . . . . .                              | 17        |
| 7.2      | Next State Learning . . . . .                              | 20        |
| 7.3      | State Prediction with Physics Loss . . . . .               | 23        |
| <b>8</b> | <b>Conclusions</b>                                         | <b>25</b> |
| 8.1      | Future Works . . . . .                                     | 26        |

# 1 Introduction

Model Predictive Control (MPC) has emerged as a powerful control strategy in robotics thanks to its ability to handle constraints and optimize system behavior over a prediction horizon. When combined with neural networks, which can learn complex system dynamics, it creates a promising framework for controlling nonlinear systems like UAVs.

This report presents the implementation and analysis of a Neural Model Predictive Control (NMPC) scheme applied to a quadrotor. We followed a bottom-up approach, testing the pipeline first on a Unicycle, which still has a nonlinear dynamics, but is easier to treat due to a simpler model and faster prototyping possibilities.

Our approach integrates Physical Informed Neural Network to approximate the quadrotor’s dynamics, which are then utilized within an MPC framework to generate optimal control inputs. We explore the effectiveness of this method in trajectory tracking tasks, comparing it with traditional control approaches.

The following sections detail the theoretical foundations, implementation specifics, and experimental results of our neural MPC system. We also discuss the challenges encountered and potential improvements for future work.

## 2 Models

Our scheme uses the MPC to generate optimal control inputs while the Neural part is used to model the dynamics and kinematics of our system. The Unicycle has been modeled with its kinematic model while for the quadrotor we have used its dynamic model.

### 2.1 Unicycle kinematic model

The kinematic model of the unicycle is a simplified representation of the movement of the vehicle in terms of position and orientation. This simple model has been derived directly from the so-called ”rolling without slipping constraint”. The state space is represented by the robot cartesian position and orientation  $[x, y, \theta]$ , while the input actions are represented by the driving and steering velocities the vehicle can assume  $[v, \omega]$ . The kinematic model of a unicycle robot it is described by the following equations:

$$\begin{cases} \dot{x}(t) = v(t) \cos(\theta(t)) \\ \dot{y}(t) = v(t) \sin(\theta(t)) \\ \dot{\theta}(t) = \omega(t) \end{cases} \quad (1)$$

## 2.2 Quadrotor dynamic model

For the quadrotor, we have used its dynamic model. We can describe this second order model by starting from the two following equations:

$$\begin{cases} m\ddot{p} = f_t + f_g + f_a + f_d \\ J\dot{\omega} + \omega \times (J\omega) = \tau \end{cases} \quad (2)$$

where the first equation, which is derived from Newton's Second Law, represents the Translational Dynamics while the second one, derived from Euler's equations, represents the Rotational Dynamics. In the first equation  $f_t$  represents the total thrust force:  $f_t = R \cdot [0 \ 0 \ T]^T$ , where  $R$  is the rotation matrix of  $(\phi, \theta, \psi)$ , the roll, pitch and yaw angles. The total thrust force  $T$  produced by the rotors is defined by the following equation:

$$T = f_1 + f_2 + f_3 + f_4, \quad \text{with} \quad f_i = k_f \omega_i^2 \quad (3)$$

where  $k_f$  is the thrust coefficient and  $\omega_i$  is the angular speed of the  $i$ -th rotor. For the Rotational Dynamics,  $\tau$  represents the torques in the body frame and can be expressed as follows:

$$\tau = \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} L(f_1 - f_3) \\ L(f_2 - f_4) \\ \tau_\psi \end{bmatrix} \quad (4)$$

where  $L$  is the distance from the center of mass of each rotor.

Given the previous equations, if we assume aereodynamics and gyroscopic effects to be negligible and that we have small roll and pitch angles, we can derive the simplified model for the quadrotor:

$$\begin{cases} \ddot{x} = -(\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) \frac{T}{m} \\ \ddot{y} = -(\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) \frac{T}{m} \\ \ddot{z} = -(\cos \theta \cos \phi) \frac{T}{m} + g \\ \ddot{\phi} = \frac{\tau_\phi}{I_x} \\ \ddot{\theta} = \frac{\tau_\theta}{I_y} \\ \ddot{\psi} = \frac{\tau_\psi}{I_z} \end{cases} \quad (5)$$

## 3 Model Predictive Control

Model Predictive Control (MPC), it is an optimal control technique where the actions are chosen solving an optimization problem, composed by a set of constraints

and an objective function to be minimized over a finite horizon.

$$\min_{u(0), u(1), \dots, u(N-1)} \sum_{k=0}^{N-1} \ell(x(k), u(k)) + \ell_f(x(N)) \quad (6)$$

$$\text{s.t. } x(k+1) = f(x(k), u(k)), \quad k = 0, \dots, N-1, \quad (7)$$

$$x(k) \in \mathcal{X}, \quad u(k) \in \mathcal{U}, \quad k = 0, \dots, N-1, \quad (8)$$

$$x(0) = x_0 \quad (9)$$

Here,  $\mathcal{U}$  and  $\mathcal{X}$  denote the admissible sets for inputs and states, respectively, while  $x_0$  is the initial state. One of the main advantages of this control approach is the ability to handle multivariable interactions and constraints. However, solving the above optimization problem in real time can be computationally challenging, especially for large-scale or highly nonlinear systems. This is one reason why neural networks have been employed to either approximate the system model or enhance the optimization process. In our case, we have used Neural Networks to predict the next state after the application of the control input given by the MPC.

### 3.1 Cost function

Our objective is to minimize at each step the error with respect to the reference trajectory, both in translational and rotational dimension, while also maintaining the module of the input actions as low as possible. A terminal state term is present too, where the translational and rotational errors of the last horizon step are tried to be minimized. Thus, the cost function is composed by the following three terms:

1. **Spatial error:** at each step of the horizon, the squared errors over the state variables are summed together

$$J_{sp_k} = W_x \sum_{i=0}^n x_{i_k}^2 \quad (10)$$

where  $x_{i_k}$  it is the  $i$ -th component of the state vector at the  $k$ -th instant.

2. **Terminal state:** to make sure the final state over the horizon is good, we add a related penalty term in the terminal state  $N$ :

$$J_{sp_N} = W_x \sum_{i=0}^n x_{i_N}^2 \quad (11)$$

where  $x_{i_k}$  it is the  $i$ -th component of the state vector at the last instant over the horizon

3. **Input:** it penalizes high magnitudes of input controls

$$J_{u_k} = W_u \sum_{i=0}^n u_{i_k}^2 \quad (12)$$

where  $u_{i_k}$  it is the  $i$ -th component of the control vector at the  $k$ -th instant.

### 3.2 Constraints

The constraints of the NLP concerns some limitations on the state of the vehicle model and on the dynamics of the system. In detail, we have just two constraints:

1. **Initial state:** the initial state of the unicycle model must be the current measured state. Therefore, defining the vehicle state as  $x$ :

$$x_0 = x_{current} \quad (13)$$

2. **Dynamics:** given the robot state at a certain time-step as  $x_t$ , an input at the same time-step as  $u_t$  and the dynamic model of the system defined as  $f_{expl}$ , we establish that the next state is obtained by applying the input at the given state according to dynamic model:

$$x_{t+1} = f_{expl}(x_t, u_t) \quad (14)$$

### 3.3 Unicycle

For our formulation, we have considered a discrete-time version of the model equations (1) obtained via Euler integration:

$$\begin{cases} x_{k+1} = x_k + T_s v_k \cos \theta_k, \\ y_{k+1} = y_k + T_s v_k \sin \theta_k, \\ \theta_{k+1} = \theta_k + T_s \omega_k. \end{cases} \quad (15)$$

According to this, the cost function terms are the following ones:

1. **Spatial error:** at each step of the horizon, the squared errors along x-axis, y-axis and error of orientation are summed together

$$J_{sp} = W_{e_x} e_x^2 + W_{e_y} e_y^2 + W_{e_\theta} e_\theta^2 \quad (16)$$

where

$$e_x = x - x_{ref} \quad (17a)$$

$$e_y = y - y_{ref} \quad (17b)$$

$$e_\theta = \theta - \theta_{ref} \quad (17c)$$

2. **Terminal state:** to make sure the final stage is a good one, we add a penalty on errors along x-axis, y-axis and error of orientation in the terminal state

$$J_{term} = W_{e_{x_N}} e_{x_N}^2 + W_{e_{y_N}} e_{y_N}^2 + W_{e_{\theta_N}} e_{\theta_N}^2 \quad (18)$$

3. **Input:** it penalizes high values of input controls

$$J_u = W_v v^2 + W_\omega \omega^2 \quad (19)$$

### 3.4 Quadrotor

Our work is based on a Crazyflie, which is a versatile open source quadrotor that only weighs 27g and fits in the palm of your hand. This particular quadrotor is characterized by a firmware that takes as inputs roll, pitch and yaw desired poses instead of control torques. This happens because there is a low level attitude controller which performs this mapping for us. As a consequence, we have used a first order dynamic model for the MPC formulation.

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = -(\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) \frac{T}{m} \\ \dot{v}_y = -(\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) \frac{T}{m} \\ \dot{v}_z = -(\cos \theta \cos \phi) \frac{T}{m} + g \\ \dot{\phi} = \frac{\phi_c - \phi}{\tau} \\ \dot{\theta} = \frac{\theta_c - \theta}{\tau} \\ \dot{\psi} = \frac{\psi_c - \psi}{\tau} \end{cases} \quad (20)$$

In this models the state is composed by the cartesian position, the cartesian velocities and the roll pitch yaw angles, while the control inputs are composed by the desired poses relative to the roll, pitch, yaw angles and the thrust.

$$\mathcal{X} = [x \ y \ z \ v_x \ v_y \ v_z \ \phi \ \theta \ \psi]^T \quad (21)$$

$$\mathcal{U} = [\phi_c \ \theta_c \ \psi_c \ T]^T \quad (22)$$

## 4 Dataset

The data the we used to train our neural models come from simulation and some real world experiments. Generally speaking, we can describe each our dataset as a tuple in the form  $\{[x_t, u_t], [x_{t+1}]\}$ , were  $x_t$  is the state of the system at time  $t$ ,  $u_t$  is the control input at time  $t$  and  $x_{t+1}$  is the state of the system at time  $t + 1$ .

For the baseline (*simple-neural-network*), no constraints on the dataset form are required, while on the physics informed models specific requirements are needed. In order to apply the PiNN framework with control, [1] has shown that the input needs to be constant in a sub-interval  $[0, T]$ . This assumption brings us to generate several trajectories with this specific constraint, and also limits the possibility of application on the data obtained in the real-world experiment. More details about this constraint are explored in the Pinn-section.

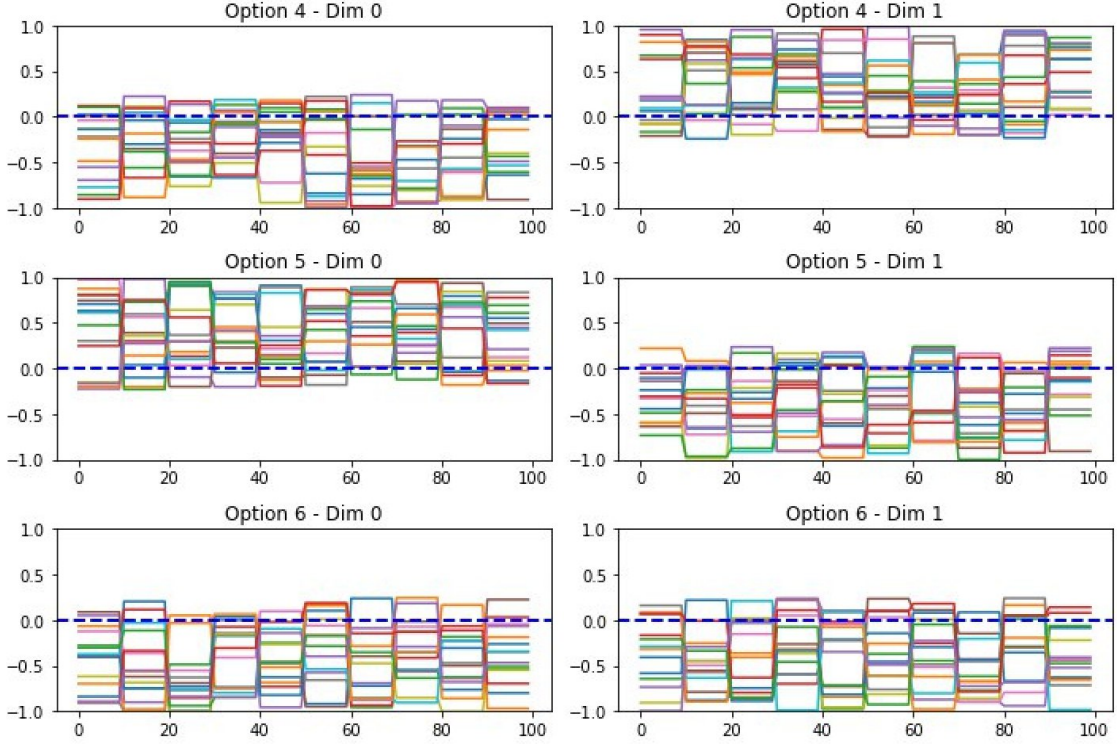


Figure 1: Example of the generated data for the unicycle (PiNN)

All the synthetic data has been obtained by integrating the system dynamics with a Runge-kutta 4th order method, while the real-world data has been extracted using the ROS interface.



When extracting data from the real-world experiments (or the gazebo simulations), state and action publishing rates were a crucial aspect to take into account. Especially, when extracting data from the real drone executing a trajectory tracking routine by using an existing MPC controller implemented by us, the action publishing rate should be high enough to make the controller safe enough. Similarly, the state publishing rate should be high, but it couldn't be too high, for communication latency reasons, which would have made the use of the real crazyflie impossible. These problems could be partially mitigated by using the gazebo simulator and setting ad-hoc publishing rates, or forcing the input to be constant (introducing a noisy approximation) for a certain amount of time in the real-world data.

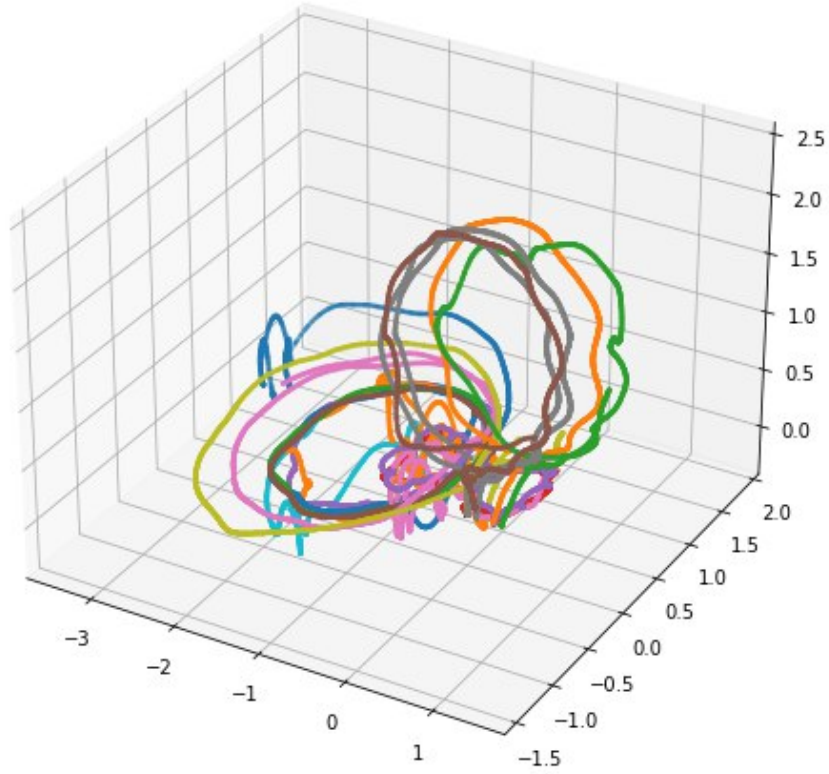


Figure 2: Example of the generated data for the quadrotor

## 5 Neural models

The model must be well-defined to be used in NMPC, in such a way to obtain reliable predictions of system behavior in response to various control actions. Neural networks are universal approximators, and they can be used both for designing the entire dynamic model of a system and approximating the disturbances and noises a system can be affected. Our purpose is to approximate the entire dynamic model of a quadrotor with a neural model, by first testing the performances on a unicycle model due to its low complexity. In this section we will start describing the used architecture used

### 5.0.1 Architecture

The general architecture of the networks is a MLP that takes as input the state of the system, the input and (in the PINN models) the time stamp. We tested the proposed approach with a various number of hidden layers and neurons, keeping the dimension of the network reasonably small for enhancing the possibility to use the on-board computer of the Quadrotor.

In the standard neural network, a smaller number of neurons was sufficient for exploit the dynamics, while the PINN framework required a larger number of neurons for smooth the loss function. Having a smoother loss landscape is crucial for let the model mimic the real dynamics.

In general, the combination of small hidden state and the absence of regularization terms in the standard models are the reason for which the general neural network framework is not able to approximate the underlying physics laws over the entire training workspace.

## 5.1 Standard neural network

### 5.1.1 Training scheme

As can be noticed in the previous section, the dataset does not contain the dynamics of the system, but just the evolution of the states. On the contrary, the forward pass of the network returns the vehicle dynamics. So, what we did is to pass the first pair  $(s_t, a_t)$  of a sample to the network; then we integrated the output in the sample time  $dt$ , and added to the previous state (which is also the input of the network). In this way we obtain the state  $s_{t+1}$ . At this point we concatenate the new obtained state with the action  $a_{t+1}$  given by the sample dataset and we pass it to the network again, and repeating this process  $k$  times we get the states evolution according to the neural model.

Ended this mechanism, the loss is computed as the mean-squared error between the neural and the nominal evolutions.

The multi-step training is a crucial step for a non-physics informed model to learn the total evolution. Experiment in a single step prediction has shown that the displacement of the vehicle is too small for having a fine-grain error measurement.

### 5.1.2 Consideration of the Dataset

As anticipated before, the dataset for such approach does not require any particular constraint. This allows us to test standard normalization techniques and so make the model more robust to different input scales. For doing so, we used a min-max normalization over the input, but we can also incorporate over the MLP architecture a normalization layer, that increases the stability of the train.

We preferred to use the layer norm in our experiment because such module does not require a preprocessing at inference time, before invoking the MPC framework. However, the normalization layer is still trained over the dataset, limiting the capacity of the network over range outside the training one.

## 5.2 Unicycle Neural model

### 5.2.1 Architecture

The neural network used to model the kinematic model of the unicycle is a simple MLP architecture, with two hidden layers of 128 neurons. The network has two inputs which are concatenated together, the current state of the unicycle (dimension 3) and the current applied actions on the vehicle (dimension 2), so it must be of dimension 5 and is represented by

$$X = [x, y, \theta, v, \omega] \quad (23)$$

The output of the network has dimension 3, and it represents the dynamics of the system, therefore it reproduces the differential equations which describe the motion of the unicycle:

$$Y = [\dot{x}, \dot{y}, \dot{\theta}] \quad (24)$$

### 5.2.2 Dataset

The dataset was generated by using the nominal model of the robot and generating 500 trajectories. Each trajectory is composed by 1000 steps, where the inputs are maintained constant for the whole episode. This choice was taken because it was noticed that the interval between consecutive time-steps is too small, therefore the

changes about the state of the unicycle are small too, and the network was not able to approximate correctly the motion of the vehicle. On the contrary, having the same input for a lot of steps allows to understand in a better way the dynamics of the system subject to specific inputs.

Each sample is not just a pair state-action at step  $t$ , but it is a window of dimension  $k$  which contains all pairs state-action from time-step  $t$  till  $t+k$  (less the last action  $a_{t+k}$  which is useless), so it is in the form

$$(s_t, a_t, s_{t+1}, a_{t+1}, \dots, s_{t+k-1}, a_{t+k-1}, s_{t+k}) \quad (25)$$

This allows to have a larger window of evolution of the system with respect to just the motion executed in one single step.

### 5.2.3 Training

As can be noticed in the previous section, the dataset does not contain the dynamics of the system, but just the evolution of the states. On the contrary, the forward pass of the network returns the vehicle dynamics. So, what we did is to pass the first pair  $(s_t, a_t)$  of a sample to the network; then we integrated the output in the sample time  $dt$ , and added to the previous state (which is also the input of the network). In this way we obtain the state  $s_{t+1}$ . At this point we concatenate the new obtained state with the action  $a_{t+1}$  given by the sample dataset and we pass it to the network again, and repeating this process  $k$  times we get the states evolution according to the neural model.

Ended this mechanism, the loss is computed as the mean-squared error between the neural and the nominal evolutions.

## 5.3 Quadrotor Neural model

### 5.3.1 Architecture

The quadrotor dynamics presents a more complex model, with a state  $x \in R^9$  and control inputs  $u \in R^4$ . For approximating such dynamics with the defined training scheme, a larger network has been tested: we have increased the number of hidden layers up to 3 and the latent dimension from 128 to 256. However, the main problem during training is related to the generation of the data.

$$\mathcal{X} = [x \ y \ z \ v_x \ v_y \ v_z \ \phi \ \theta \ \psi]^T \quad (26)$$

$$\mathcal{U} = [\phi_c \ \theta_c \ \psi_c \ T]^T \quad (27)$$

$$\mathcal{Y} = [\dot{x} \ \dot{y} \ \dot{z} \ \dot{v}_x \ \dot{v}_y \ \dot{v}_z \ \dot{\phi} \ \dot{\theta} \ \dot{\psi}]^T \quad (28)$$

### 5.3.2 Dataset

Since our quadrotor implementation is based on the Crazyflie, we have generated data according to the first order model used in the MPC formulation (20). The data have been extracted from ROS messages given directly by the robot and the simulation. We have defined 19 different types of trajectories that belong to 3 main categories:

- horizontal\_circle
- vertical\_circle
- lemniscate

As previously stated, the dataset collection, both in simulation and on the real robot is characterized by some limitations that directly affect the training procedure.

### 5.3.3 Training

Like we have done in the unicycle’s case, the input is composed by the current state of the quadrotor concatenated with the control inputs and the current timestamp, while the output is composed by the following state of the robot. The loss is computed by taking the mean squared error between the nominal evolutions and the predicted ones.

## 6 Physical Informed Models

### 6.1 Physics-Informed Neural Network: Introduction

Traditional methods for solving differential equations, such as finite element or finite difference methods, often require discretization of the domain and can be computationally expensive for high-dimensional problems. Physics Informed Neural Networks (PiNNs) propose an alternative approach where the solution is approximated by a neural network that is trained not only on data but also by enforcing the underlying physical laws via the differential equations. This approach should be able to generalize outside the training data, fully respect the underlying physics. With outside

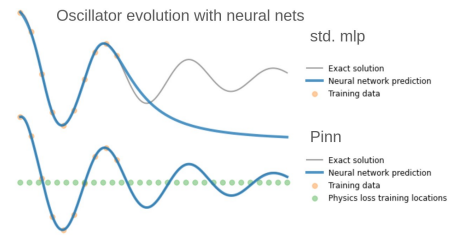


Figure 3: Example of the Pinn behavior.

training data, we intend to say that the model generalize for sample that has never been shown **inside** the range of training.

The main loss component of the Pinn is called Physics Loss, and minimize the discrepancy between the dynamics of the system and the derivative of the network itself.

The standard PiNN models takes in input just the time  $t$  at which we want to predict the evolution of the system. So, the output of the model is define as  $x(t)$ . For impose particular constraints, other losses are taken into account. Example of such loss are the so call *Boundary loss* and the *Initial loss*. The first one is used to impose the boundary condition of the system, while the second one is used to impose the initial condition of the system.

One limitation of such approach is that the model is not able to generalize to different initial condition: the model need to be trained again for predict a new evolution. Also, as few works have shown, the classica formulation of the Pinn is not stable over long time interval. Finally, the standard pinn model is not able to include the control input into the dynamics.

For this reason, we have used a extended version of the framework, namely, the PiNN-C (*PiNN with control* [1]). Such works shown as incorporate the control input on the prediction scheme, and has became the de-facto standard for model non-autonomus dynamic evolution ([2] [3]). The Pinn-C model is analyze in the next section of this report.

## 6.2 Physics-Informed Neural Network with Control

The Pinn-c framework rely on the assumption that the input is constant over a sub-interval  $[0, T_k]$ . This assumption is used for ensure that the gradient of the output is not "corrupted" by the derivative of the state respect the input signal. In this framework, the input of the network is composed by the initial condition of the sub-transition, the time stamp for which we want to obtain the prediction and the constant control input. Doing so, we obtain a set of tuple of the form  $\{[x_0, \bar{u}_k, t_k], [x_{t_k}]\}$ , where  $x_0$  is the initial state of the system,  $t_k$  is the time stamp for which we want to predict the evolution and  $\bar{u}_k$  is the control input applied to the system.

The pinn-c, in the ground, just model a set of smaller evolution, and the discriminative factor remains the time input, as in the standard formulation.

For concatenate the sub-trajectories, the starting continion of the next sub-trajectory is setted as the output of the previous interval at time  $T$ . For a concise representation, see figure 4 Finally, while the standard pinn can not rely on the trajectories from a dataset, the pinn-c must use it. This dependency is modeled as a imitation

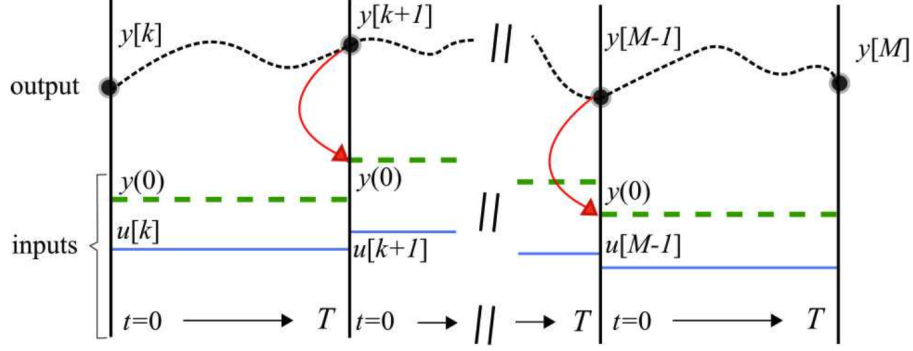


Figure 4: Pinn-c sub-interval integration

loss, similar to the standard MLP case.

In such framework it's not possible to rely on multi-step integration for calculate the loss term. Due the definition of input/output of the network, it's impossible to concatenate the previous integration into the future input. Extra details on the loss are given on the section below.

### 6.2.1 Dataset requirements, generation and constraints

In our experiment, after generate the evolution of the system, we notice that a good coverage of the input space of the network is required. In the first phase of the experiment, we have generated a set of trajectories using a fixed family of input, like sinusoidal or step function. Although good result in the training and in the evaluation of the model (isolated form the mpc), we have notice that several phenomena conflict with the integration of the network into the mpc. In particular, the network was not able to predict the evolution of the system when the control input was null. Also, basic motion like forward or backward trajectories remains challenging.

For this reason, we have decided to generate a new dataset, where the control input is randomly generated but all the possible combination of sing of the input are taken into account: we have generated 8 different combination for the Unicycle:

- $v > 0, w > 0$
- $v > 0, w < 0$
- $v < 0, w > 0$
- $v < 0, w < 0$
- $v > 0, w = 0$
- $v < 0, w = 0$
- $v = 0, w > 0$
- $v = 0, w < 0$

The last possible combination,  $v = 0, w = 0$ , has been excluded from the dataset,

as we model a additional loss term that relies on sampling. The dataset has been generated using a Runge-Kutta 4th order method, and the sub-interval has been set to 0.1 seconds. In order to approximate the real evolution of the system, the PiNN-C framework needs different sample in which the input is constant. For this reason, after having discretize the input signal, we have taken up to 10 sub-evolution, leading to a reading of the trajectory of 0.02 seconds.

Under this assumption and constraint, we have seen that the previously mentioned phenomena are no longer present or at least has been minimized under a significant threshold.

### 6.2.2 Loss definition and training scheme

The overall loss function  $\mathcal{L}$  used is composed by the sum of two main terms, which are the data loss (also call *imit loss*)  $\mathcal{L}_{data}$  and the physics loss  $\mathcal{L}_{phy}$ .

$$\mathcal{L}(\theta) = \mathcal{L}_{data}(\theta) + \mathcal{L}_{phy}(\theta), \quad (29)$$

The first term defines the discrepancy between the prediction of the network and the observed data, while the second one specifies the correct modeling of the differential equation. Usually  $\mathcal{L}_{data}$  is define as the mean squared error between the prediction and the ground truth, while  $\mathcal{L}_{phy}$  is defined using the residual of the differential equation.

For a correct implementation of the physics loss, a sampling strategy is required. The usage of the physics loss can be seen as a self-supervised learning instance, due to the fact that it does not rely on any dataset.

At each iteration, the learner must sample a set of initial states, input and time from the relative distribution. Such samples are called "particles". A great number of sampled particle is important for a correct evaluation of the loss function, as reported in the self-supervised learning literature. For such particle, we don't rely on the prediction of the network, but only on the gradient with respect to the input. Thanks to automatic differentiation, the gradient of the network can be easily calculated.

After having obtained the gradient, we can also calculate the output of the differential equation, and then calculate the loss term:

$$\mathcal{L}(\theta)_{physics} = \frac{1}{N} \sum_{i=1}^N (\nabla_x - ode(x, u, t))^2 \quad (30)$$

In the previous equation,  $ode$  is the differential equation that we are interested to learn as underlying prior, while  $\nabla_x(X)$  represents the gradient of the network with respect to the input state. The  $ode$  function is the same differential equation that we have introduced in the first part, and describes the evolution of our target



systems.

An extra term has been added, as a new boundary condition. When the model has 0 input, the network must become an identity function, returning the same input state as output. This loss is defined as the mean squared error between the prediction and the input. Such application is possible only in the case of driftless system, as the unicycle. In the case of quadrotor, we prefer to include sample with 0 input in the dataset and rely on the integration of the dynamic itself for incorporate such prior. This approach result in a less robust model, due the fact that the network only see few example of such evolution. In the case of driftless system, such loss can be implemented using a sampling strategy. As in the physical loss, the input vector has been sampled in the correct range, then we manually zero the input element.

---

**Algorithm 1** Training Loop: Loss calculation

---

- 1: **Input:**  $X_{\text{data}}, y$
  - 2: **Initialize:**  $L \leftarrow 0$
  - 3: **Compute Predictions:**  $\hat{y} \leftarrow \mathcal{M}(X_{\text{data}})$
  - 4: **Update Loss with MSE:**  $L \leftarrow L + \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
  - 5: **Sample Input Data:**  $X \sim \mathcal{D}$
  - 6: **Calculate Gradient:**  $J \leftarrow \nabla_X \theta$
  - 7: **Compute Analytical ODE:**  $\text{ode}(X) = f(X)$
  - 8: **Update Loss with ODE Term:**  $L \leftarrow L + \frac{1}{n} \sum_{i=1}^n (\text{ode}(X) - \nabla_X \theta)^2$
  - 9: **Sample a Point:**  $X_{\text{point}} \sim \mathcal{D}$
  - 10: **Zero the Input:**  $X_{\text{input}} \leftarrow 0$
  - 11: **Update Loss with Boundary Term:**  $L \leftarrow L + \frac{1}{n} \sum_{i=1}^n (X_i - \hat{y}_i)^2$
  - 12: **Output:** Updated loss  $L$
- 

## 7 Results

In this section we will report some results obtained by training the neural model and applying it to an MPC based trajectory tracking problem applied to the unicycle robot.

### 7.1 Derivative Learning

In figure 5 we can observe that the loss converges pretty quickly.

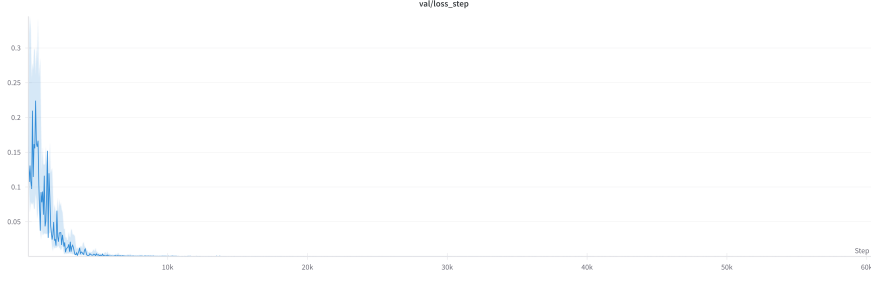


Figure 5: Loss plot for the derivative prediction task

This reflects also in a successful evaluation on the test set, where predictions are very close to the nominal model, with very few exceptions. Also the integration in the MPC is successful, achieving nearly identical results to the nominal model.

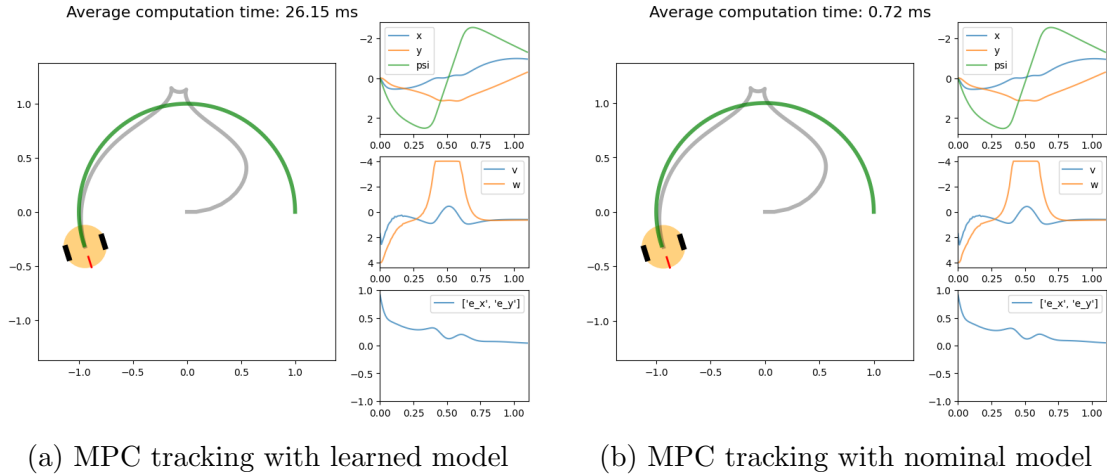


Figure 6: Derivative prediction evaluation

In these plots, the MPC simulation has a discretization timestep of 50 ms, in order to make the neural-mpc controller feasible, since it has higher computational demands than the nominal model. The neural model is also potentially more scalable, since it can be trained on a variety of more complex models, while potentially keeping the same computational cost. The nominal model performs better in computational terms, but the neural model is potentially more robust to disturbances and model errors, especially if we were to inject noise in it during the training process. This discrepancy between computational performance is also imputable to the integration method used by the mpc solver in Acados. Indeed, the neural network that predicts the next state instead of the derivative will perform better inside the MPC.

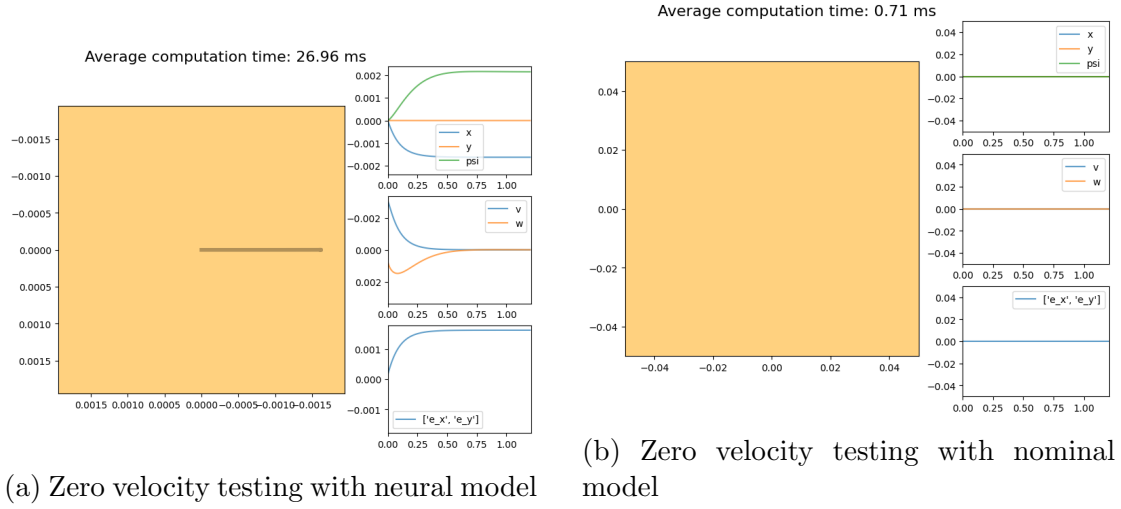


Figure 7: Derivative prediction evaluation

In figure 7 we can see that the neural model has some issues when the desired tracking velocity is zero. The neural mpc has a imperceptible drift, while the nominal model is, of course, completely still. This derives from the intrinsic data-related nature of neural model, which can't captivate the differential equations profoundly.

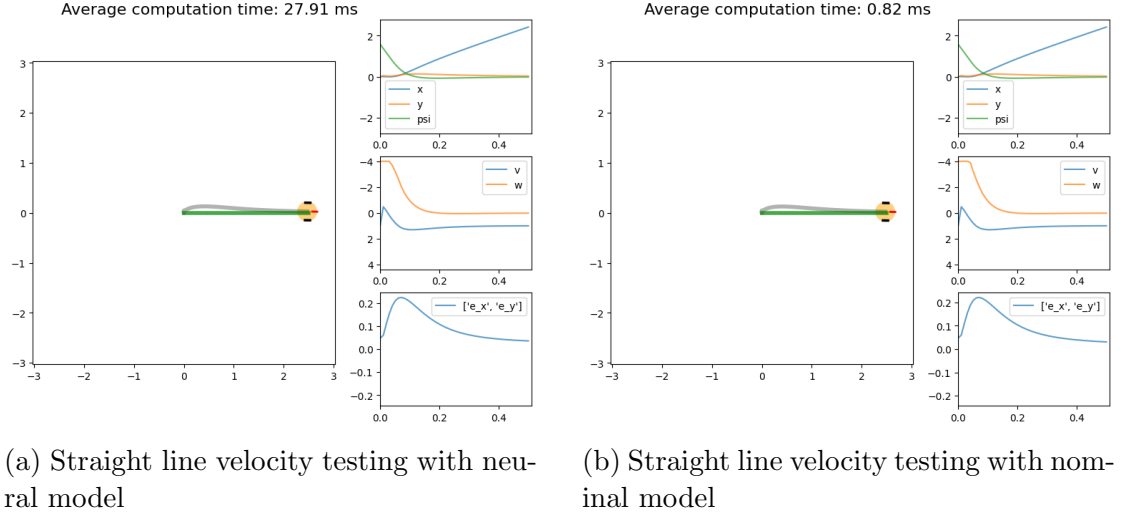


Figure 8: Derivative prediction evaluation

In figure 8 it is even clearer how the neural model actually perfectly mimicks the nominal model, having the same error profile.

## 7.2 Next State Learning

The loss for the state prediction task is depicted in figure 9. The trend is very similar to before.

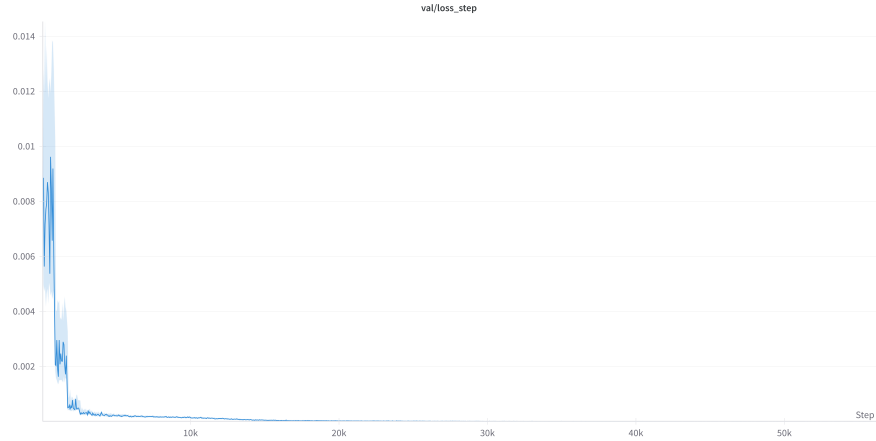


Figure 9: Loss plot for the state prediction task

Similarly to before, this reflects positively in the evaluation on the test, but there are some edge cases where predictions are not as accurate as before. This probably comes from the fact that we use Euler integration to propagate the nominal system dynamics. Or the cause could be also that the network is overfitting on the training data without any notion on the actual system dynamics. Thus, when seeing new unseen data, even if still in the same distribution, generalization capabilities are not excellent. We tried to mitigate this problem by adding some callbacks in the training loop that stop the training when the validation loss stops decreasing, but this low generalization power behavior remains.

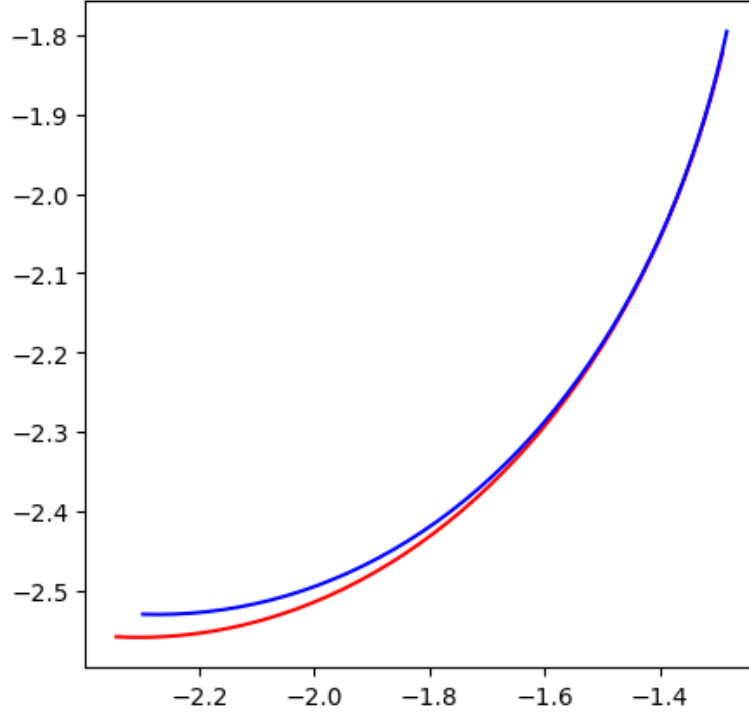


Figure 10: unsuccessful evaluation on a test sample

This poor generalization capabilities reflect also negatively on the MPC tracking task. In figure 11 we can see that the

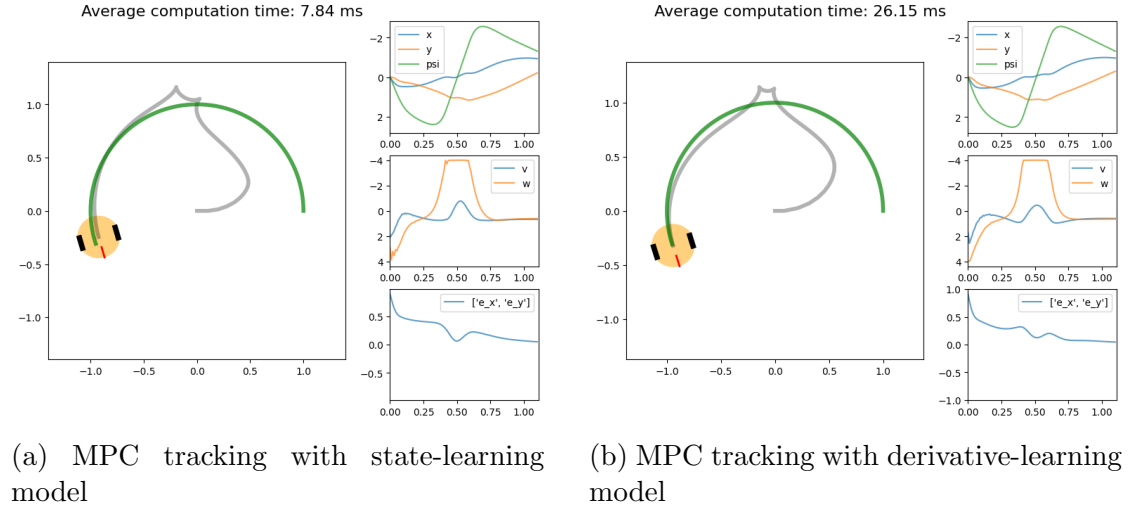


Figure 11: State prediction evaluation on a circle

As we can see in the figure above, the state-predicting neural model has some issues to track the trajectory with high precision. The tracking error descends more slowly than with the derivative-predicting neural model, and the final error is slightly higher.

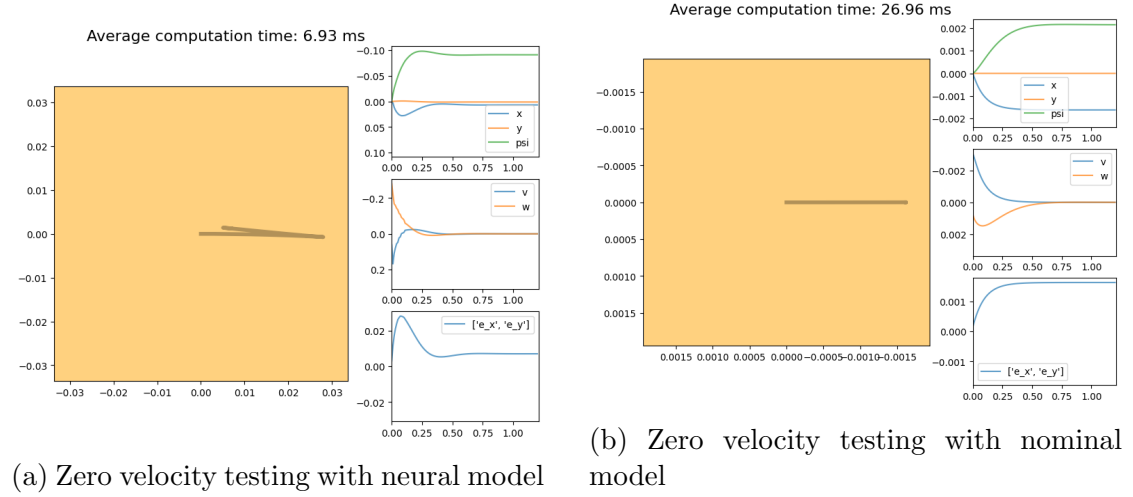


Figure 12: Derivative prediction evaluation

Also the zero velocity testing is not as successful as with the derivative neural model, but interestingly, after a small initial drift, the unicycle goes back to its desired position and stays still, as desired.

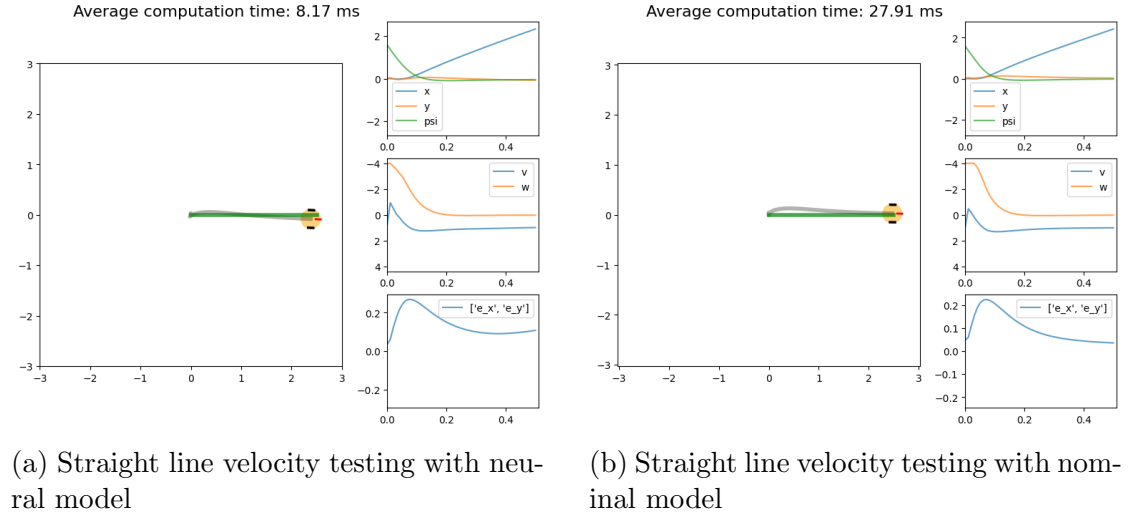


Figure 13: Derivative prediction evaluation

In figure 13 the slight drift is vividly visible, as we can see from the error curve, which does not descend monotonically.

### 7.3 State Prediction with Physics Loss

Adding the physics loss makes the mpc tracking task slightly better, as we can see in figure 14. Even computational performance is slightly better, suggesting that the MPC solver struggled less than before to find good solutions. The error profile shows a steeper descent when the learner is augmented with the physics loss.

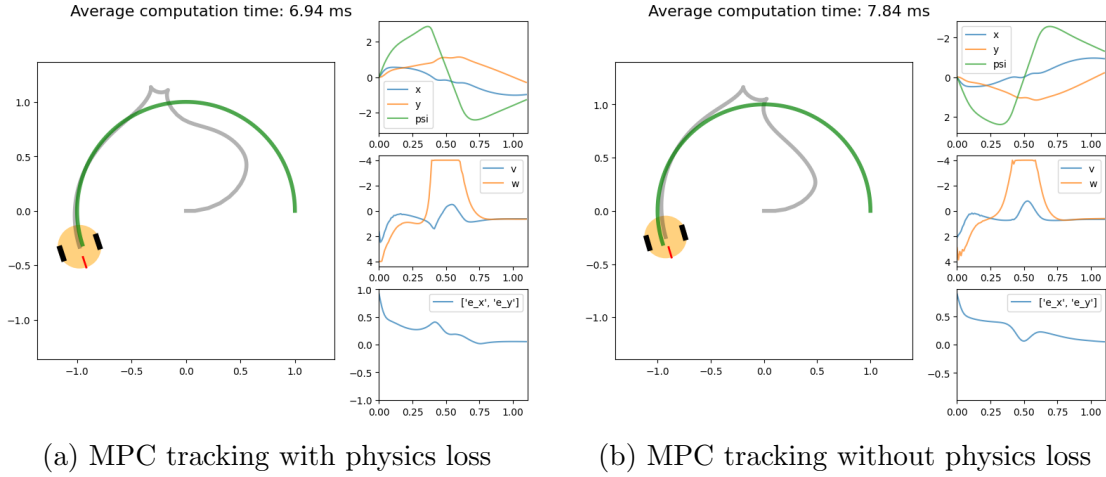
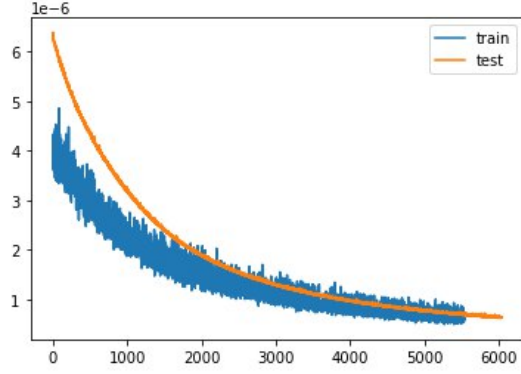
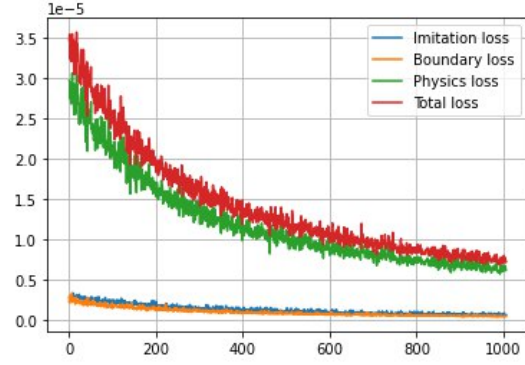


Figure 14: Pinn prediction evaluation

In figure 17 the difference is even more evident. On the left side, the neural model with physics loss is able to track the trajectory, with a monotonically descending error, while on the right side this is not the case. In figure 15a we can see the training and test loss for the imit component, while on the figure 15b we can see the effect of the single component of the loss.



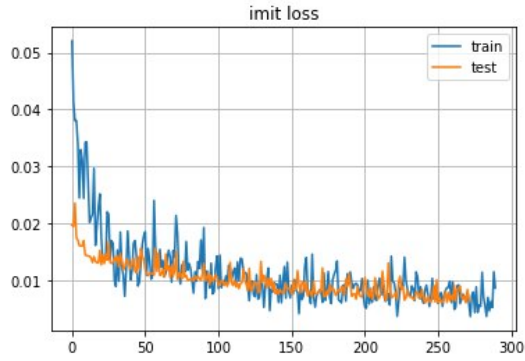
(a) test and train loss for the imit component



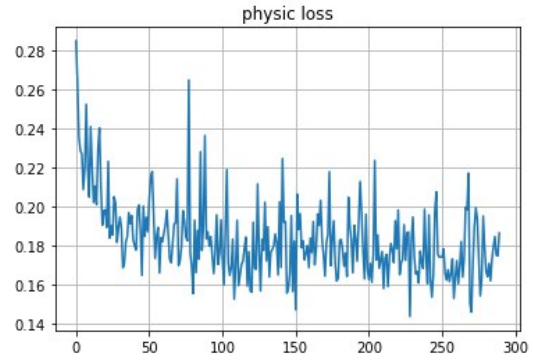
(b) train loss for the pinn-c model

Figure 15: Pinn loss evaluation, unicycle

In comparison, we can also see the loss plot for the quadrotor model (figure: 16). We can see that the loss are noise, and the reached value is several order of magnitude bigger than the unicycle model:



(a) test and train loss for the imit component of the quadrotor model



(b) physics loss of the pinn-c model for the quadrotor

Figure 16: Pinn loss evaluation, quadrotor



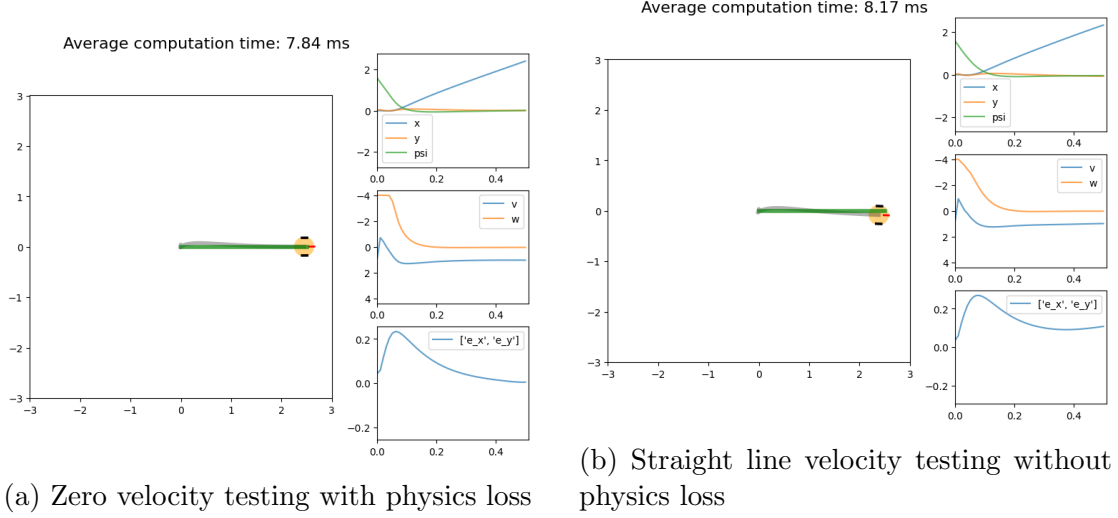


Figure 17: Derivative prediction evaluation

## 8 Conclusions

In this project we wanted to explore the possibility to integrate Pytorch-learned models within CasADI and Acados to provide a learning based model predictive controller. This approach allows to exploit the vast adaptability of neural networks while exerting some control over the system through MPC, which allows to impose constraints. We have shown that this approach is feasible, while presenting also some limitations. By far, the main limitation is given by the intrinsic nature of neural networks, which don't take into account the actual physical meaning of the equations governing the dynamical system.

This can be partially solved by using a physics-informed neural network, which we did for the unicycle. While the neural approach shows promising results, in leveraging internal disturbances and improving controller accuracy, the standard MPC framework retains advantages in scenarios with limited or incomplete data distributions. Moreover, the dimensionality challenge — where increasing the state/input space dimensions necessitates exponentially more sampled points — remains a significant hurdle.

Despite these challenges, the results underscore the potential of physics-informed models to address complex control problems. These models offer a unique capacity to incorporate domain-specific knowledge, paving the way for improved adaptability and robustness in controller design. However, the development of more specialized and robust constraints is essential to maximize their applicability and ensure consistent performance across diverse scenarios.

Another problem is given by how the neural network is integrated within the

MPC. The optimization problem has its own computational graph, and the neural network has its own too, so these two graphs need to be merged. This problem was solved by using L4Casadi.

Of course, this is just an exploratory project and many problems remain to be solved.

## 8.1 Future Works

We are confident that the physics informed model must be a future direction for the research, while better and more general for of constraint could be implemented. Other option for future improvement are the combination of dynamic systems and neural models, in order to predict more stable evolution over undefined range. An example of such integration could be the usage of reservoir computing: few works has been demonstrate that such approach can be used for deal with unbalance and non-homogeneous input signal, while the output layers can be tuned in a physics informed fashion.

Other problems and consideration are related to the sim2real transfer: Considering the single basin hypothesis, a double stage training mechanism could be used for linearly connect multiple models. Keeping as fixed the model obtained in nominal condition, and interpolate other network trained on different dataset, must results in a upgrade of the main model. Our conjecture is that the pinn could constraint the network to have single basin, that is the global minima for the main model, while other models trained on noisy dataset could contribute for reach a more general solution.

Also, a moving average technique in the training phase could be beneficial for the convergence of the network. Future works are required for integrate the Neural-MPC framework into real world application.

We hope that this project can offer a comprehensive overview of the challenge and possibility for future step, as all the challenges encountered during the development are current state of the art research problems.

## References

- [1] Eric Aislan Antonelo, Eduardo Camponogara, Laio Oriel Seman, Jean Panaioti Jordanou, Eduardo Rehbein de Souza, and Jomi Fred Hübner. Physics-informed neural nets for control of dynamical systems. *Neurocomputing*, 579:127419, April 2024.
- [2] Jonas Nicodemus, Jonas Kneifl, Jörg Fehr, and Benjamin Unger. Physics-informed neural networks-based model predictive control for multi-link manipulators, 2021.

- [3] Sourav Sanyal and Kaushik Roy. Ramp-net: A robust adaptive mpc for quadrotors via physics-informed neural network, 2023.